

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10: 4.1 STATISTICS I

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Madam Achieng, the Deputy Principal of a day secondary school in Kisumu, has given the Grade 10 class the test scores of the same 20 learners in Subject 1 (Mr. Otieno) and Subject 2 (Mrs. Wanjiku). The pass mark is 50 out of 100. Answer every question in the space under it, using the data from our lessons. Show your working for every calculation. Write grouped-data answers as estimates and give reasons in full sentences. When you talk about WHY the scores differ, say what we cannot yet know and use words such as 'may', 'could' and 'we need more data'. Your answers should help you answer the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps?

Part 1: Organising the scores into frequency tables

Here are the scores in ascending order.

Subject 1: 67, 69, 70, 72, 73, 74, 76, 77, 78, 80, 81, 82, 84, 85, 86, 88, 89, 90, 91, 95.

Subject 2: 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, 42, 44, 45, 51, 58, 60, 60.

(a) Using a common class width of 10, with classes written 10-19, 20-29, ... up to 90-99, make one grouped frequency table for both subjects. Use tally marks grouped in fives and show the frequency for each subject. Check your totals.

(b) Why did we use the same classes for both subjects?

(c) If you made an ungrouped frequency table for Subject 1, what would you notice, and why does grouping help Madam Achieng more? (Key Inquiry 1: How do we organise data using frequency distribution tables?)

Write your answer here:

Part 2: Finding the typical score

Use the ordered lists in Part 1 (the sum of the Subject 1 scores is 1607 and the sum of the Subject 2 scores is 677).

(a) Calculate the mean and the median of each subject.

(b) State the mode of each subject, and explain what it means when a set has no mode or two modes.

(c) Compare the typical score in each subject with the pass mark of 50. What does this suggest about whether the learners are weak overall?

Write your answer here:

Part 3: Reading position and spread from cumulative frequency

The grouped frequencies are:

Class	Subject 1 f	Subject 2 f
10-19	0	4
20-29	0	4
30-39	0	5
40-49	0	3
50-59	0	2
60-69	2	2
70-79	7	0
80-89	8	0
90-99	3	0

- (a) Make a cumulative frequency table for Subject 2 using the upper class boundaries (less-than values).
- (b) Using the 5th, 10th and 15th learners, estimate Q1, the median and Q3 for Subject 2, and its interquartile range (IQR). Show your working.
- (c) For Subject 1 the estimates are $Q1 = 73.8$ and $Q3 = 87.0$. Find its estimated IQR and compare it with that of Subject 2. What does this tell you about the spread of the middle half of the learners in each subject?
- (d) Why are these values called estimates?

Write your answer here:

Part 4: Claim, evidence and reasoning

A classmate says: 'These 20 learners are all weak, so Madam Achieng should tell the whole class to repeat the term.'

(a) Write a claim-evidence-reasoning response. Use at least three pieces of evidence from the tables, measures or graphs (for example pass counts, a measure of centre and a measure of spread).

(b) We saw that Subject 1 histograms and polygons are narrow and to the right, and Subject 2 histograms and polygons are wide and to the left. Explain what this shape tells you that the raw lists did not.

Write your answer here:

Part 5: Recommendations and why data helps decisions

Madam Achieng wants a final report that answers the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps?

(a) Write one evidence-based recommendation each for Mr. Otieno, for Mrs. Wanjiku and the learners, and for the 16 learners who scored below 50 in Subject 2. Link each to the data.

(b) Name two reasons that could explain the difference in scores. Explain why you cannot yet claim that either is the cause, and what further data you would collect.

(c) Answer Key Inquiry 2: Why is it important to collect, organise and interpret data in making decisions? Use what changed between the raw lists and your tables, measures and graphs.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Organising data (frequency tables, Key Inquiry 1)	Builds a complete, correct grouped table with a common class width of 10, correct tallies and frequencies totalling 20 for each subject, and clearly explains why grouping is more useful than an ungrouped table.	Builds a mostly correct table with one small error, and gives a basic reason for using the same classes or for grouping.	Table has several errors or totals that do not equal 20, or gives no reason for the choice of classes.
Calculating measures (mean, median, mode, quartiles, IQR)	All measures are correct (80.35, 80.5, 33.85, 31.5, modes 32 and 60, IQR estimates 13.2 and 24.2) with clear	Most measures are correct with working shown, with a small slip, or the reason for 'estimate' is incomplete.	Several measures are incorrect or working is missing, and does not recognise grouped values as estimates.

	working, and grouped values are called estimates with a reason given.		
Interpreting tables, measures and graphs	Uses centre, spread, shape and pass rate together to compare the subjects, and explains what the graphs show that the raw lists hid.	Uses at least two features (such as the mean and the pass count) to compare the subjects, with limited reference to spread or shape.	Only lists numbers or describes one subject, with no comparison.
Claim-evidence-reasoning and recommendations	Gives a clear claim supported by several correct pieces of evidence, and recommendations for Mr. Otieno, Mrs. Wanjiku, the learners and the 16 learners below 50, each linked to the data.	Gives a claim with some evidence and recommendations for most groups, but the link between the data and the recommendation is sometimes weak.	Claim is unsupported or recommendations are missing or not linked to the data.
Separating what the data shows from what it cannot explain	Treats reasons for the score difference as hypotheses using words such as 'may' and 'could', and names suitable extra data (topic-by-topic marks, a second test).	Recognises that causes are not known but gives an unclear or incomplete plan for finding out.	States a cause for the score difference as a fact.

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